Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python lists:

```
p = [[1, 2], [3, 4]]
q = ['a']
```

What is the result of evaluating the expression below?

```
3 * q + p
```

A. `['a', 'a', 'a', [1, 2], [3, 4]]`

B. `[['a'], ['a'], ['a'], [1, 2], [3, 4]]`

C. `['aaa', [1, 2], [3, 4]]`

D. `['a', [1, 2], 'a', [3, 4]]`

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python code:

```
l = [1, 2, 3]

def fun():
    l = [9, 9, 9]

fun()
print(l)
```

What will be the output of this program?

A. `[9, 9, 9]`

B. `[1, 2, 3]`

C. `[1, 2]`

D. `Error`

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the list:

```
l = [100, 200, 300, 400, 500]
```

What will be the output of the following expression?

```
print(l[4:-10:-2])
```

A. `[500, 300]`

B. `[500]`

C. `[500, 400, 300]`

D. `[]`

Q #4 **Multiple Choice Type** **Award: 1** **Penalty: 0.33** **Programming in Python**

Consider the following Python code:

```
L = []
for w in ["AB", "CD"]:
    L += w
print(L)
```

What will be the output of this program?

A. `['AB', 'CD']`

B. `['A', 'B', 'C', 'D']`

C. `[['A','B'], ['C','D']]`

D. Error

Q #5 **Multiple Choice Type** **Award: 1** **Penalty: 0.33** **Programming in Python**

What is the value of the following expression?

```
len(set(['A', 'b', 'b', 'a']))
```

A. 2

B. 3

C. 4

D. 0

Q #6 **Multiple Choice Type** **Award: 2** **Penalty: 0.67** **Programming in Python**

Consider the following Python code:

```
def size(hierarchy, head):

    if head not in hierarchy:

        return 1

    total = 1

    for emp in hierarchy[head]:

        total += size(hierarchy, emp)

    return total



hierarchy = {

    "CEO": ["A", "B"],

    "A": ["C", "D"],

    "C": ["E"],

    "B": ["F"]

}



print(size(hierarchy, "CEO"))
```

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python code:

```
A = [[10, [20, 30]], 40]

B = A

A[0][1][0] = 99

A[1] = 50

A[0][0] = 77

print(B)
```

What will be the output of the program?

A. `[[10, [20, 30]], 40]`

B. `[[77, [99, 30]], 50]`

C. `[[77, [20, 30]], 50]`

D. `[[10, [99, 30]], 40]`

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python code:

```
A = [[1, 2], [3, 4]]

B = A[:]

A[0][1] = 99

A[1] = [7, 8]

print(B)
```

What will be the output of B?

A. `[[1, 2], [3, 4]]`

B. `[[1, 99], [3, 4]]`

C. `[[1, 99], [7, 8]]`

D. `[[1, 2], [7, 8]]`

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python function:

```
def partition_like(D, s1, s2):
    if s1 >= s2:
        return
    if D[s1] > D[s2]:
        D[s1], D[s2] = D[s2], D[s1]
    partition_like(D, s1 + 1, s2 - 1)

D = [9, 1, 7, 3, 4, 8]
partition_like(D, 0, 5)
print(D)
```

What will be the output of this program?

A. `[9, 1, 7, 3, 4, 8]`

B. `[8, 1, 7, 3, 4, 9]`

C. `[8, 1, 3, 7, 4, 9]`

D. `[9, 1, 3, 7, 4, 8]`

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python function:

```
def prefix_min(X):
    m = X[0]
    out = [m]
    for i in range(1, len(X)):
        m = min(m, X[i])
        out.append(m)
    return out


X = [5, 2, 8, 1, 7]
print(prefix_min(X))
```

What will be the output of the above code?

A. [5, 2, 2, 1, 1]

B. [5, 2, 2, 1, 7]

C. [5, 5, 5, 1, 1]

D. [5, 2, 8, 1, 1]

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in Python

Which of the following code snippets will print the output $(5, 6, 7)$?

A.
```
x,y,z = 5, 6, 7
print((x, y, z))
```

B.
```
def f(*t):
    print(t)
f(*(5, 6, 7))
```

C.
```
a, *b = (5, 6, 7)
print(b)
```

D.
```
def f(a, * b) :
    print((a, b))
f(5, 6, 7)
```

Q #12 Multiple Choice Type Award: 2 Penalty: 0.67 Programming in Python

Consider the following Python code snippet:

```
def bar(item, D={}):
    if item not in D:
        D[item] = 1
    else:
        D[item] += 1
    print(D)

for i in ["x", "y", "x"]:
    bar(i)
```

What will be the output of this program?

A.
```
{'x': 1}
{'x': 1, 'y': 1}
{'x': 2, 'y': 1}
```

B.
```
{'x': 1}
{'y': 1}
{'x': 1}
```

C.
```
{'x': 1}
{'x': 1, 'y': 1}
{'x': 2, 'y': 1, 'x': 1}
```

D. An error occurs because a default dictionary cannot be modified.

Q #13 Multiple Choice Type Award: 2 Penalty: 0.67 Programming in Python

Given the list:

```
words = ["a", "cat", "python", "hi", "book", "sun"]
```

Which of the following list comprehensions returns all words with length greater than 3?

A. `[w for w in words if len(w) > 3]`

B. `[len(w) > 3 for w in words]`

C. `[w if len(w) > 3 else "" for w in words]`

D. `[w for w in words if w > 3]`

Q #14 | **Multiple Choice Type** | **Award: 2** | **Penalty: 0.67** | **Programming in Python**

Consider the following Python code:

```
A = {"red", "blue"}
B = {"blue", "green"}
C = {"green", "yellow"}

if "blue" in A:
    A = A - B

if "green" in B:
    B = B | C

if "yellow" in C:
    C = C - A

print(A, B, C)
```

Which of the following is the correct output?

A. `{'red'} {'yellow', 'blue', 'green'} {'yellow', 'green'}`

B. `{'blue'} {'green'} {'yellow'}`

C. `{'red', 'blue'} {'green', 'yellow'} {'green'}`

D. `{'blue'} {'blue', 'yellow'} {'yellow', 'red'}`

Q #15 | **Multiple Choice Type** | **Award: 2** | **Penalty: 0.67** | **Programming in Python**

What is the value of the following Python expression?

```
''.join('h a n d s'.split())
```

A. `['h', 'a', 'n', 'd', 's']`

B. `'h a n d s'`

C. `'hands'`

D. None